HERO HOUSING FINTECH CHALLENGE · PROBLEM STATEMENT 2 · TEAM SOLUTION REPORT

Nirnay: a humanless underwriting engine for Home Loans and LAP

Nirnay (निर्णय, "decision") takes a Home Loan or Loan Against Property application from the first tap to an Approve, Refer or Reject decision. Clean cases need no person. Every check runs as code, every rule points to the regulation it comes from, and an auditor can replay every decision.

Version 1.0 · 5 November 2026 · Written for our team · "Nirnay" is a working name

8 of 8 required points covered, each mapped to a module	**2 channels** mobile app and web portal on one engine	**3 outcomes** Approve, Refer or Reject, always with reasons
Under 2 s to compute a decision once data is in (design target)		

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The short version

Problem Statement 2 asks for an engine that underwrites Home Loans and LAP with no person in the decision path. It has to follow NHB/RBI rules, internal credit policy, KYC/AML law, the Fair Practices Code and audit standards. Nirnay is our answer.

- Customers apply on a **mobile app** or the **web portal**. Consent screens let the engine pull data from credit bureaus, Account Aggregators, UIDAI, GSTN, EPFO, state land records and CERSAI.

- The engine runs **eight automated gates**: identity, AML, fraud and documents, policy rules, property, affordability, risk grade and authority limit. It returns **Approve, Refer or Reject** with reason codes.
- A **compliance gate** blocks any decision that breaks a coded regulation. Every decision is saved as a tamper-evident **decision record** that can be replayed years later.

Six choices that make it work

Two-stage decision

An in-principle offer in minutes from the customer's own data, and a final sanction once property checks finish. Builder projects we have already checked skip most property work.

Rules and AI kept apart

Rules are versioned tables the credit team edits. Models produce scores. A large language model (LLM) reads documents and drafts text, and never sets an outcome.

Compliance as code

Each regulation becomes a rule tagged with its source clause. If a rule fails, no decision goes out.

Refer is a feature

Cases the rules say need a person, such as a politically exposed customer, go to an officer with a ready file. Officer decisions train the next model.

Replay any decision

The same inputs with the same rule and model versions always give the same answer, so audits are exact.

One engine, many front doors

App, portal, partner app and branch staff call the same APIs, so the same rules apply everywhere.

How to read this report. In five minutes: read this section and the flow in section 5. If you are building a part: read both architecture sections, then find your role in the build plan. You can test the decision logic yourself in the simulator.

The problem in plain words

Underwriting is the lender's answer to three questions. Can this person repay? Will they repay? Is the property good security if they don't? For a Home Loan the property is being bought. For LAP the customer already owns it and borrows against it.

Today a file moves through many hands. Sales collects papers. Operations checks KYC. Credit reads salary slips and bank statements. A lawyer checks the title. A valuer visits the property. Each team re-checks what the last one did, files go back to the customer for missing papers, and the customer waits days.

Problem Statement 2 asks us to replace that chain with one engine that decides on its own, follows every rule, and can show why it decided.

What “humanless” means in our design

- No person is needed to approve or reject a clean file.
- People still own the rules, the models and the hard cases. The engine sends a case to a person (**Refer**) when a regulation requires it, for example a politically exposed person, or when data is missing or uncertain.
- A customer can always ask for a person to review a rejection through the grievance process.

Why this is hard

- **Data is scattered.** Bureaus, banks, income tax, GST, PF and land records in every state all hold a piece.
- **Property is the slowest part.** Land records and encumbrance certificates are online in many states, not all, and formats differ.
- **Self-employed income is hard to read.** Many LAP customers have real income that is poorly documented.
- **Fraud targets every step.** Fake salary slips, edited bank statements, one property mortgaged twice, agents and valuers working together.
- **Rules change.** RBI consolidated its directions on 28 November 2025 and keeps amending them.
- **Every decision needs an explanation** that the customer, the auditor and the supervisor can each follow.

A note on “NHB regulations”. Since 9 August 2019 the RBI regulates housing finance companies, and the National Housing Bank (NHB) supervises them through inspections and regulatory returns. Most rules that used to be NHB directions now sit in RBI's Housing Finance Companies Directions, consolidated again on 28 November 2025. In this report, “NHB/RBI rules” means both.

Home Loan and LAP are different problems

	Home Loan (HL)	Loan Against Property (LAP)
Purpose	Buy, build or improve a home	Any lawful need: business, education, medical, other
Property	The home being bought, often under construction	A property the customer already owns
Typical borrower	Salaried and self-employed	Mostly self-employed and small-business owners
Loan-to-value (LTV) limit	RBI tiers: 90% up to ₹30 lakh, 80% above ₹30 lakh up to ₹75 lakh, 75% above ₹75 lakh	Set by internal policy; we assume 60% (illustrative)
Income proof	Salary slips, Form 16, bank credits, EPFO	ITR, GST returns, bank cash-flow

	Home Loan (HL)	Loan Against Property (LAP)
Extra checks	Builder and RERA status, source of own contribution, sale agreement	End use, existing charges, self-occupied or rented, business vintage
Main fraud risks	Fake income, inflated price, fake builder or seller	Property mortgaged twice, inflated value, round-tripped turnover

How we cover all eight points

The challenge rules say to cover every point on the “What you need to do” page. This table maps each one to the part of Nirnay that handles it.

#	Required point	What we built	Key pieces
1	Integrate data sources	Integration hub with one adapter per source, behind a consent manager; data-flow diagram	AA, 4 bureaus, UIDAI, CKYCR, GSTN, EPFO, land records, CERSAI, RERA
2	Eligibility and policy	Real-time eligibility maths plus policy rules as versioned decision tables, separate sets for HL and LAP	Rules engine, policy studio, maker-checker
3	Fraud detection and KYC/AML	Automated KYC, AML screening that clears false matches by itself, fraud engine using rules, ML and a link graph	eKYC, PAN, face match, liveness, watch-lists, graph analytics
4	Risk scoring	Readable scorecard (live) and a boosted-tree challenger give a 300–900 score and grade A–E, with points per factor	Scorecard, LightGBM, SHAP, calibration
5	Documents and property	Document AI and property checks (title, encumbrance, CERSAI, RERA, valuation model, geo-tagged photos) in the same flow	OCR, tamper checks, AVM, LLM deed reader
6	Compliance checks	Compliance gate: each regulation is a coded rule with its source; decision blocked unless all pass	RBI HFC, KYC and Digital Lending Directions, FPC, PMLA, DPDP
7	Instant decisioning	Eight-gate pipeline, under 2 s to compute; Approve (with counter-offer), Refer or Reject	Orchestrator, decision service
8	Explainability and governance	Reason codes, hash-chained decision records, replay, model governance mapped to RBI's FREE-AI framework	Audit ledger, audit portal, model cards

One engine, two front doors: app and web portal

Customers, partners and staff all reach the same engine through the same APIs. The **mobile app** is for doing things on the move with a camera. The **web portal** is for longer work at a desk, and it is where staff, vendors and auditors work.

Mobile app (Android and iOS)

Customer app

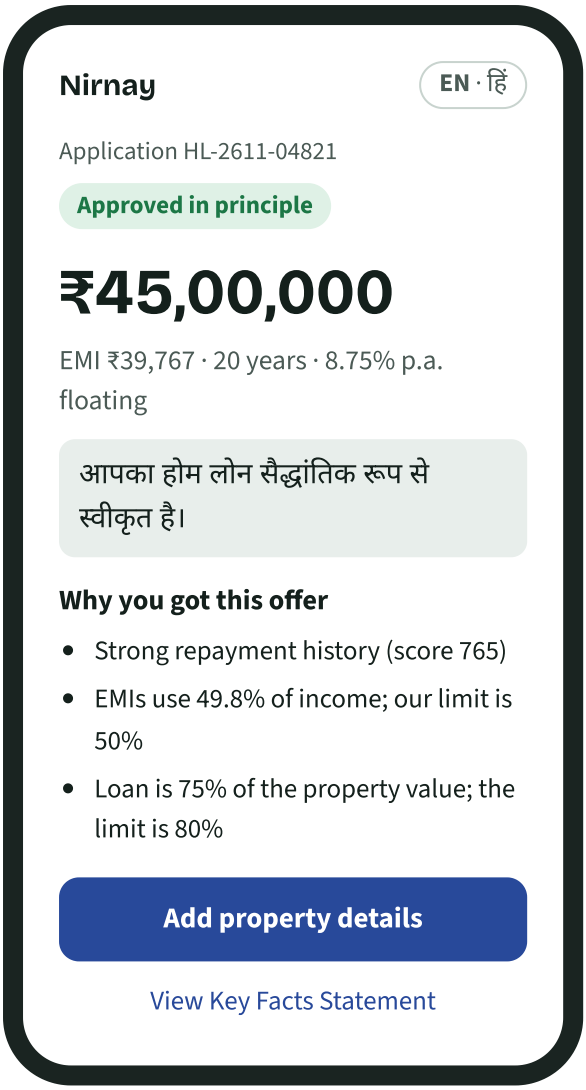
- Sign in with mobile OTP; the device is bound to the account.
- Auto-fill from PAN, Aadhaar eKYC or DigiLocker, so the customer types very little.
- Plain-language consent screens for the bureau pull, Account Aggregator and the DPDP privacy notice.
- Camera: selfie with liveness, document capture with blur and glare checks, geo-tagged property photos and a short video walk-through.
- Quick eligibility and EMI options before the full application.
- Decision screen with reasons in English, Hindi and other Indian languages.
- Key Facts Statement, eSign, and eNACH or UPI AutoPay inside the app.
- Live status, WhatsApp and SMS updates, a help assistant, and one-tap grievance with officer details.

Partner app (DSA, connector, builder sales)

- Start a lead; the customer gives consent on their own phone, so the partner never sees OTPs.
- See pending items and payouts per file.
- Builder sales can look up units in pre-approved projects.

Web portal

- **Customer portal:** the same journey as the app on a laptop; invite a co-applicant; download letters.
- **Partner portal:** bulk leads; builders upload RERA papers, approvals and plans once to get a project pre-approved.
- **Credit console:** the Refer queue with an AI-drafted file note, evidence viewer, re-pull of data, a decision with a mandatory reason, SLA timers, and a second approver for large loans.
- **Vendor portal:** empanelled lawyers and valuers receive referred cases, upload reports, and see SLAs.
- **Compliance and audit portal:** search and replay any decision, see rule and model versions, export evidence, regulatory reports, Fair Practices metrics such as turnaround times, rejection reasons and complaints.
- **Policy studio:** edit LTV, FOIR and score cut-offs as tables; maker-checker approval; test on past cases before going live; effective dates.
- **Model governance console:** model inventory, drift and fairness dashboards, override rates, and a kill switch that sends all new cases to Refer.



Credit console · Refer queue — 4 open · sample data

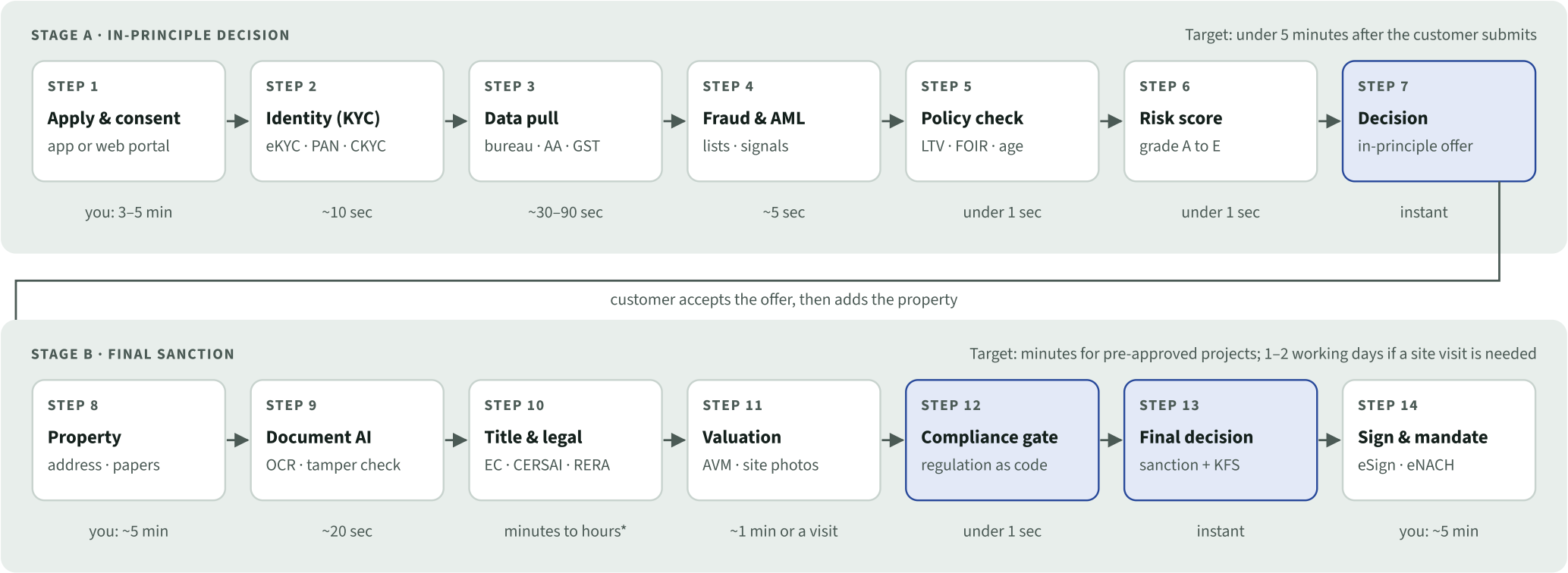
Application	Amount	Why referred	Gate	
HL-2611-04807	₹38.0 L	Possible PEP match on name and date of birth	AML	3
LAP-2611-01133	₹72.0 L	Title chain has a gap: one co-owner deceased, no mutation online	Property	6
HL-2611-04815	₹1.24 Cr	Above the ₹1 crore auto-sanction limit	Authority	1
HL-2611-04819	₹22.5 L	No credit history; bank cash-flow is steady for 24 months	Policy	5

Each row opens a pre-filled file: facts with links to evidence, the engine's draft offer, and the exact rule that referred it. The officer's decision and reason are saved to the decision record.

Sample screens. The app numbers come from the worked example in Requirement 2; the queue rows are made up.

End-to-end flow: application to decision

The flow has two stages. Stage A uses data about the person and ends in an in-principle offer. Stage B adds the property and ends in the final sanction. Both stages end in the same three outcomes.



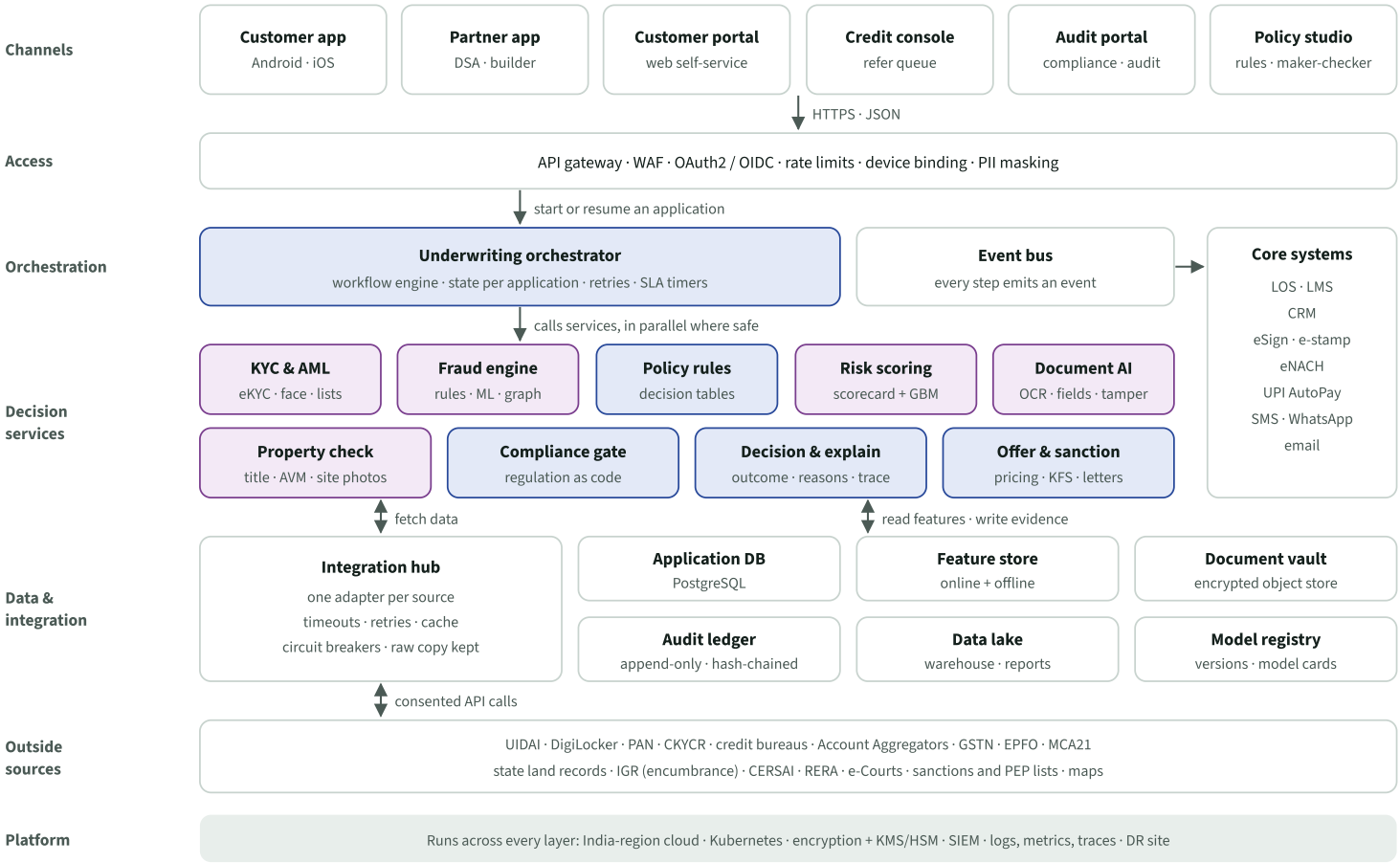
The application moves through 14 automated steps in two stages. Times under each step are design targets. *Title checks depend on whether the state's land records and encumbrance certificates are online; where they are not, the case is referred to an empanelled lawyer through the vendor portal.

The engine itself decides in under two seconds once its inputs are in. Time in the flow goes to waiting for data: the customer filling the form, the Account Aggregator returning statements, a state portal returning an encumbrance certificate. If a source is slow, the application waits in a clear status the customer can see. The engine never guesses missing data.

Pre-approved projects. A builder's project is checked once for title, approvals and RERA registration, then marked pre-approved. A unit in that project only needs unit-level checks, so Stage B can finish in minutes and many Home Loans complete in one sitting.

Solution architecture (SA)

Seven layers. Requests come in through the channels and pass the gateway. The orchestrator runs each application as a state machine and calls the decision services in order. Services get outside data through the integration hub and write evidence to the data layer. Every step emits an event that the audit ledger, notifications and dashboards consume.



service that uses AI models rules-based (deterministic) service or core engine part infrastructure, data or channel

The orchestrator is the only part that knows where an application stands. Services each do one job, so a rule change in policy does not redeploy fraud or scoring.

What each layer does

Channels	Customer and partner apps, and the web portals for customers, partners, credit officers, auditors and policy owners. All are thin clients over the same APIs.
Access and security	API gateway with a web application firewall, OAuth2/OpenID Connect login, role-based access, rate limits, device binding, and masking of personal data in logs.
Orchestration	A durable workflow engine keeps one state machine per application: which steps ran, what they returned, what is pending, and SLA timers. Steps are idempotent, so a retried bureau call does not create a second enquiry on the customer's report.
Decision services	Nine services with one job each. AI-backed ones return scores and facts; rules-based ones apply policy and regulation. The decision service combines their results into one outcome.
Data and integration	The integration hub talks to outside sources. The application database holds state, the feature store serves model inputs, the vault keeps documents, and the append-only audit ledger keeps decision records.
Core systems	The loan origination and management systems, CRM, eSign and e-stamp, eNACH and UPI AutoPay mandates, and notifications receive events after sanction.
Platform	India-region hosting, Kubernetes, encryption with managed keys, a security monitoring (SIEM) feed, full observability, and a disaster-recovery site.

Technology stack

Layer	Prototype (hackathon)	Production option	Why
Mobile app	Flutter	Flutter	One codebase for Android and iOS; good camera support
Web portal	React + Next.js	React + Next.js	Role-based screens, server rendering for speed
Gateway	Kong or NGINX	Kong or cloud API gateway + WAF	Auth, rate limits, routing in one place
Services	Python FastAPI	FastAPI for AI services, Java Spring Boot for core	Python fits ML; Java fits long-lived core services
Workflow	Simple state machine, or Temporal	Temporal or Camunda 8	Durable state, retries, timers, full history
Rules	json-rules-engine or DMN tables	DMN (Camunda) or Drools	Business-editable tables with versions
Data	PostgreSQL, Redis	PostgreSQL (high availability), Redis	Reliable, well understood

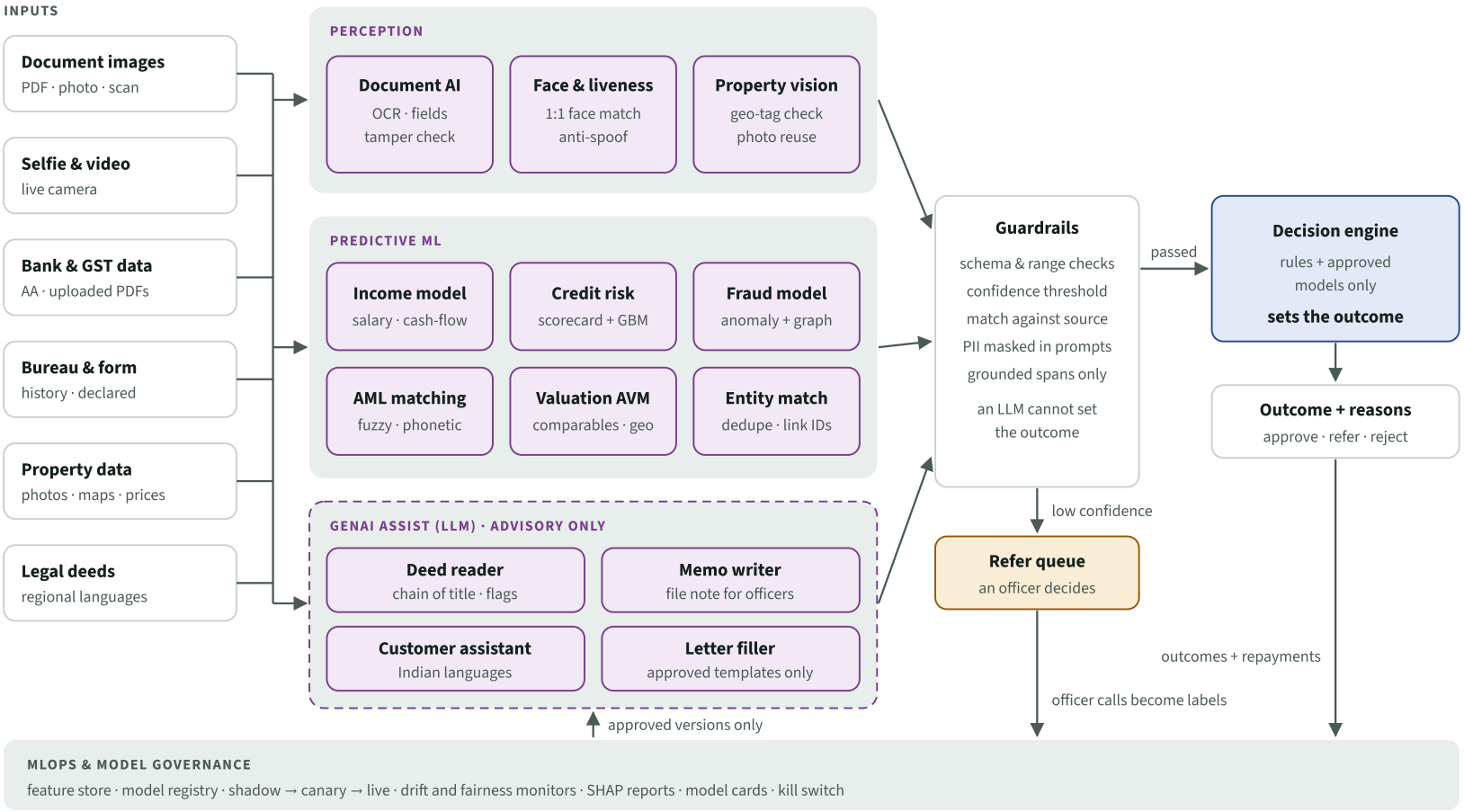
Layer	Prototype (hackathon)	Production option	Why
Events	Redpanda	Kafka	Replayable event log for audit and analytics
ML	scikit-learn, LightGBM, SHAP, MLflow	Same + Feast feature store + drift monitoring	Explainable models with version control
Document AI	PaddleOCR or Tesseract + layout rules	OCR and layout models hosted in India	Indian documents, many scripts
LLM	Hosted model through a private endpoint	India-region private endpoint, no data retention	Data residency and privacy
Fraud graph	NetworkX	Neo4j	Finds rings through shared phones, devices, accounts
Storage	MinIO (S3-compatible)	Object store with write-once (object lock)	Tamper-evident evidence and records
Observability	OpenTelemetry, Prometheus, Grafana	Same + SIEM	Traces show where time goes in each application

Non-functional targets

Quality	Design target
Decision compute time	Under 2 seconds for 95% of decisions, once inputs are present
Stage A end to end	Under 5 minutes after the customer submits
Availability	99.9% for the decision path; a source outage pauses only the affected step
Recovery	Recovery point under 15 minutes, recovery time under 2 hours
Data residency	All personal and loan data stored and processed in India
Audit	100% of decisions have a record; 100% of replays match the original
Security	Encryption in transit and at rest, mTLS between services, secrets in a vault, least-privilege roles

AI solution architecture (AI SA)

AI does three jobs in Nirnay. It **reads** documents, photos and deeds. It **predicts** income, default risk, fraud and property value. It **writes** file notes and help answers. It does not decide. The decision engine sets every outcome from rules and approved models, after each AI output has passed the guardrails.



Every AI output passes through guardrails before the decision engine sees it. The GenAI group sits inside a dashed boundary because its output is advisory: it extracts facts and drafts text, and it can never set an outcome. Officer decisions on referred cases become training labels.

AI components

Component	Technique	Input → output	Guardrail
Document AI	OCR and layout model, document classifier, field extractor; tamper checks on file metadata, fonts, pixel patterns and QR codes	Images or PDFs → typed fields with a confidence for each	Low confidence asks for a re-upload; every field is cross-checked with a source API where one exists
Face and liveness	Face embeddings for 1:1 match; passive liveness to catch photos of photos and masks	Selfie and ID photo → match and liveness scores	Thresholds set from false-accept tests; a fail offers video KYC
Property vision	GPS and timestamp checks, perceptual hashing to catch reused photos, photo-versus-satellite comparison, construction-stage estimate	Photos and video → location and authenticity flags	Any mismatch sends a valuer
Income model	Text classification of bank narrations; salary, EMI and bounce detection; cash-flow volatility	Account Aggregator data or PDF statement → monthly income, obligations, red flags	Shown next to declared income; a big gap goes to Refer
Credit risk	Logistic scorecard with weight-of-evidence points (live); gradient-boosted trees with monotonic constraints (challenger); calibration to probability of default	Features → score, grade, points per factor	Independent validation before use; monthly stability and fairness checks
Fraud model	Supervised model on known fraud, anomaly detection, and graph analytics over shared phones, addresses, devices, accounts, agents and valuers	Application and its links → fraud score, ring alerts	Flags only; declaring a fraud follows RBI's fraud process
AML matching	Fuzzy and phonetic matching tuned for Indian names, transliteration between scripts, scoring on secondary IDs	Name, date of birth, IDs → match score	Possible matches go to Refer; the customer is never told why
Entity match	Record linkage across applications and internal data	Records → one linked customer, duplicate flags	Conservative thresholds; uncertain links are not merged
Valuation (AVM)	Gradient boosting or hedonic regression on registered prices, circle rates, listings and location features	Property → value range and confidence	Engine uses the lower of AVM and valuer value; low confidence sends a valuer

Component	Technique	Input → output	Guardrail
GenAI assist (LLM)	India-hosted LLM with retrieval over approved documents	Deeds → facts with source spans; reason codes → letter from an approved template; referred case → file note; customer question → answer	Each extracted fact must point to text in the source; no promises to customers; prompts logged; personal data masked

Six rules we follow for AI

1. Outcomes come from rules and validated models. An LLM never decides.
2. Every model has an owner, a model card, a version and a validation report before it goes live.
3. No protected attributes as inputs (gender, religion, caste, marital status). We also test for proxies such as PIN code.
4. Every AI output carries a confidence. Below the threshold, the case goes to Refer.
5. Models and data stay in India-region infrastructure. Personal data is masked in LLM prompts, and providers may not train on our data.
6. A kill switch sends every new case to Refer if monitoring shows a problem.

Model lifecycle (MLOps)

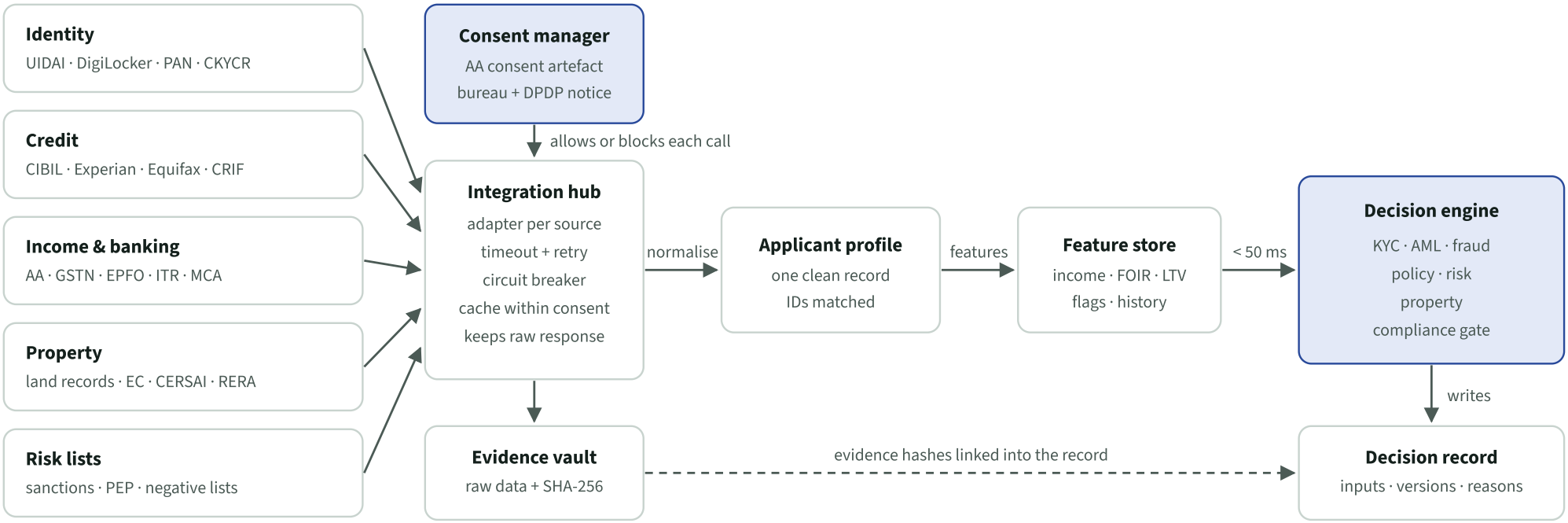
1. **Data:** versioned training sets built from the feature store, so any model can be rebuilt exactly.
2. **Train:** scorecard and challenger trained on the same data split.
3. **Validate:** an independent reviewer checks accuracy (Gini, KS), stability, fairness and explanations.
4. **Approve:** the model risk committee signs off; the model card goes to the registry.
5. **Shadow:** the model scores live cases without affecting decisions; we compare it with the live model.
6. **Canary:** a small share of traffic uses the new model, with automatic rollback.
7. **Live and monitored:** drift (population stability index above 0.25 triggers review), performance on matured loans, approval and refer rates by segment, override rates, latency.
8. **Retrain:** on a schedule or when a monitor fires, back to step 1.

Cold start. We do not have Hero's past loan data in the hackathon. Our demo model trains on the public Home Credit Default Risk dataset and we will say so in the demo. In production the scorecard is built on Hero's own loan book. Until there is enough history, it runs on bureau score plus an expert scorecard.

1 REQUIREMENT 1 OF 8

Integrate data sources

Asked: connect to multiple external data sources and show how data flows into the engine.



Data never reaches the engine without passing the consent manager and the integration hub. The raw response is hashed and kept, so an auditor can check that the engine saw exactly what the source sent.

Sources we connect

Source	What we get	How we connect	Used for
UIDAI (Aadhaar)	eKYC data and photo, or offline XML	Licensed KUA partner where permitted; otherwise Aadhaar offline XML or DigiLocker	Identity, address
DigiLocker	Issuer-signed Aadhaar, PAN, driving licence and other issued documents	Partner API with customer consent	Identity, document authenticity
PAN verification (Protean, UTIITSL)	Name, status, Aadhaar–PAN link	API	Identity, duplicate checks
CKYCR (run by CERSAI)	Existing KYC record; we upload new ones	API	Faster KYC
Credit bureaus: TransUnion CIBIL, Experian, Equifax, CRIF High Mark	Score, loans, days past due, recent enquiries	API with consent; idempotent calls	Policy rules, risk score
Account Aggregator network	Bank statements and other financial data from participating institutions	As a Financial Information User, with a consent artefact per request	Income, cash-flow, obligations
GSTN	GST returns and turnover	Through a GST Suvidha Provider or AA, with consent	Self-employed income (mainly LAP)
Income tax records (ITR, AIS, Form 26AS)	Declared income, tax deducted	Customer-consented fetch or upload	Income check
EPFO (UAN)	Employer, PF contributions	Partner API with consent	Employment proof
MCA21	Company and director data	API	Employer and business checks
State land records and IGR	Owner, survey details, encumbrance certificate, registered deeds	State portals and APIs; coverage varies by state	Title, ownership
CERSAI security-interest registry	Existing mortgage on the property	Search API	Catching a property mortgaged twice
RERA (state authorities)	Project registration, builder, timelines	Portal	Under-construction Home Loans
e-Courts	Cases by party name	Search	Litigation on the borrower or property

Source	What we get	How we connect	Used for
Circle rates, maps, satellite (for example ISRO Bhuvan)	Guidance values, location, land use	Public data and map APIs	Valuation, location risk
Sanctions and PEP lists	UN Security Council list, lists under the UAPA, PEP and adverse-media data	Daily list refresh	AML screening
Internal systems (LOS, LMS, CRM)	Past loans and applications, internal negative list	Internal API	Duplicates, past fraud, existing exposure

After the decision, the same hub connects to **execution services**: Aadhaar eSign, e-stamping, eNACH and UPI AutoPay mandates, bank account verification (penny drop), SMS and WhatsApp, and the loan systems.

How the hub keeps data reliable

- **Consent first.** No call goes out without a valid consent for that purpose; consents are stored with their artefact IDs.
- **One adapter per source.** Each adapter turns the source's format into our standard format and has its own timeout and retry rules.
- **Fallbacks.** If the Account Aggregator is down, the customer can upload a PDF statement and Document AI reads it. The engine marks which path was used.
- **Circuit breakers.** A failing source is paused so it does not slow every application; affected steps wait in a clear status.
- **Evidence kept.** The raw response is stored with a SHA-256 hash, and the decision record points to it.
- **Quality checks.** Names, dates and amounts are normalised, and records from different sources are matched on PAN, mobile and date of birth.

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2 REQUIREMENT 2 OF 8

Eligibility and policy

> **Asked:** assess eligibility in real time for Home Loans and LAP, and validate each case against internal credit policies.

Credit policy lives in the engine as **decision tables**, not hard-wired code. The credit team edits them in the policy studio. Every change needs a second approver (maker-checker), carries a version and an effective date, and is tested against past applications before it goes live. Home Loans and LAP have separate rule sets.

How the eligible amount is worked out

```
Max EMI = net monthly income × FOIR cap - existing EMIs
Loan by income = Max EMI × [1 - (1 + r)^-n] ÷ r (r = monthly rate, n = months)
Loan by property = LTV cap × property value
Eligible loan = the lowest of: loan by income, loan by property, product limit, a
```

Worked example: Ravi, 32, salaried, buying a ₹60 lakh flat

- Net income ₹1,00,000 a month; existing EMI ₹10,000; asks ₹45 lakh for 20 years at 8.75%.
- FOIR cap 50% → max EMI = ₹50,000 − ₹10,000 = **₹40,000**.
- At 8.75% for 240 months, each ₹1 of EMI supports about ₹113.16 of loan → loan by income ≈ **₹45.26 lakh**.
- A loan between ₹30 lakh and ₹75 lakh has an 80% LTV cap → loan by property = **₹48 lakh**.
- Eligible = lowest of ₹45.26 lakh, ₹48 lakh and the ₹45 lakh asked → **₹45 lakh**, EMI **₹39,767**. Total EMIs are 49.8% of income and LTV is 75%.

Policy parameters

Parameter	Home Loan	LAP
Minimum age at application	21	21
Maximum age when the loan ends	65 salaried, 70 self-employed	65 salaried, 70 self-employed
Maximum tenure	30 years	15 years
Bureau score	700+ passes; 650–699 Refer; below 650 Reject; no history Refer	
Recent late payments (12 months)	30+ days late Refer; 90+ days late Reject	
FOIR cap by monthly income	45% up to ₹50,000; 50% up to ₹1,50,000; 60% above	
LTV cap	RBI tiers: 90% / 80% / 75%	60%
Auto-sanction limit	₹1 crore; above this, Refer to a senior officer	
Interest rate used in the maths	8.75% floating	10.50% floating

Only the HL LTV tiers come from regulation. Every other value is illustrative. Hero's real credit policy plugs in through the policy studio.

Product-specific checks

Home Loan

- Source of the customer's own contribution traced in bank data.
- Builder and project in the pre-approved list, or RERA-registered with a clean record.
- Disbursal schedule tied to construction stage for under-construction homes.
- Stamp duty and registration are left out of the property cost for LTV, except where RBI allows them for homes costing ₹10 lakh or less.

LAP

- End-use declaration captured; end-use monitoring set up after disbursal.
- Property owned by the applicant or co-applicant; no undisclosed charge.
- Business vintage and income from GST and ITR agree with bank credits.
- Existing loan on the property must be taken over or closed at disbursal.

Real time. All rules run in under a second once features are ready. A quick check runs on declared data before the full application, so the customer sees a realistic amount early.

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3 REQUIREMENT 3 OF 8

Fraud detection and KYC/AML

Asked: detect fraud and meet KYC/AML requirements without manual review.

KYC, fully digital

1. **Mobile OTP and device binding.** The device fingerprint is saved for fraud checks.
2. **PAN check.** Name and status from the PAN service, plus the Aadhaar–PAN link status.
3. **Aadhaar.** eKYC through a licensed partner where Hero is permitted, or Aadhaar offline XML or DigiLocker. Aadhaar numbers are stored only as tokens in an Aadhaar Data Vault, as UIDAI requires.
4. **CKYCR.** Search and download an existing record; upload new records after onboarding.
5. **Face match and liveness.** The selfie is matched to the ID photo; passive liveness rejects photos of photos and masks.

6 **Fallback.** If the face match fails, the customer is offered video KYC (V-CIP) with an official, as RBI's KYC Directions allow.

AML, without manual review for clear cases

- **Screening:** names and dates of birth are checked against the UN Security Council lists, the lists under the UAPA, PEP lists and adverse media, at onboarding and again whenever a list updates.
- **Smart matching:** fuzzy, phonetic and cross-script matching handles Indian name spellings, so “Mohammed” and “Muhammad” or “Choudhary” and “Chaudhari” are compared properly.
- **Auto-clearing:** a name hit is cleared automatically when date of birth, PAN, father's name or nationality clearly differ. Only real possible matches reach a person.
- **Risk category:** each customer is rated low, medium or high from occupation, geography, product and source of funds. High risk gets enhanced checks.
- **Rules that need a person:** a politically exposed person needs enhanced checks and senior management approval under the KYC Directions, so the engine refers these cases by design.
- **No tipping-off:** the customer never sees an AML reason. Their message says only that the application needs more review. Suspicious cases are prepared for the Principal Officer, who files with FIU-IND.

Fraud checks

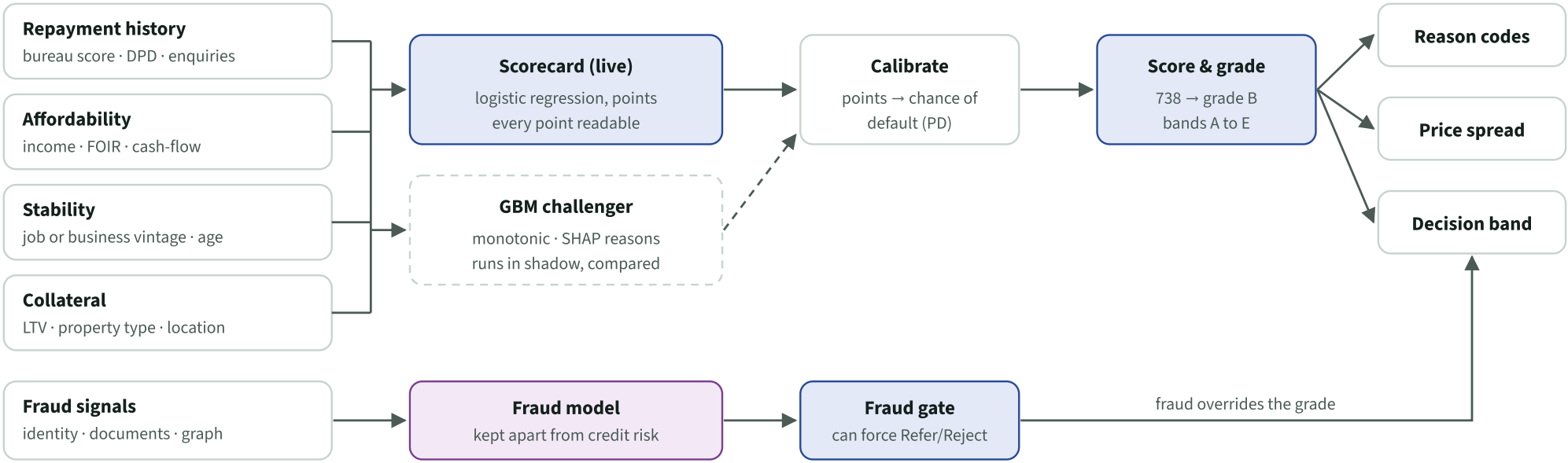
Fraud type	Signal	Automated check
Identity theft, synthetic identity	Face mismatch, PAN name mismatch, a device seen on many applications	Face match, PAN API, device fingerprint, velocity rules
Inflated income	Salary slip higher than bank credits, EPFO or AIS	Cross-source income reconciliation; employer check via EPFO and MCA
Forged documents	Edited PDF metadata, mixed fonts, wrong formats, QR that does not match	Document AI tamper checks, issuer verification through DigiLocker or QR
Fake bank statements	Balances that do not carry forward, odd transaction patterns	Prefer AA data (cannot be edited); balance-continuity checks on PDFs
Property fraud	Same property mortgaged twice, seller not the owner, inflated value	CERSAI search, encumbrance certificate, land-record owner match, AVM versus declared price, geo-tagged photos
Collusion rings	Many applications sharing a phone, address, employer, account, agent or valuer	Graph analytics with community detection
Round-tripping	Money moving in circles to inflate turnover	Transaction graph on bank and GST data

High-confidence fraud is rejected; suspected fraud is referred. The engine raises flags and early warning signals only. Classifying a borrower as a fraud follows RBI's Fraud Risk Management Directions, which require a fair process such as a show-cause notice before that label is applied.

4 REQUIREMENT 4 OF 8

Risk scoring

Asked: score the risk of every application and show how the score is produced.



The live score comes from a scorecard where every point can be read and explained. The boosted-tree model runs in shadow as a challenger. Fraud is scored separately so that a strong credit profile can never hide a fraud signal.

How the score is produced

1. **Features:** the feature store gives about 40 inputs in four groups: repayment history, affordability, stability and collateral.
2. **Points:** the scorecard gives each input points. For Ravi: base 650, repayment history +33, no missed payments +10, affordability +10, collateral +15, salaried stability +20.
3. **Total and calibration:** points add up to 738. Calibration on past loans maps each score to a probability of default.
4. **Grade:** 738 falls in grade B. The grade sets the decision band and the price spread under Hero's board-approved interest-rate policy.
5. **Reasons:** the factors that lost the most points become the reason codes. In the challenger, SHAP values play the same role.

Grade	Score	Decision	Price spread (illustrative)
A	760–900	Approve	Base rate
B	700–759	Approve	Base + 0.25%
C	650–699	Refer	Set by the officer within policy
D	600–649	Reject	None
E	300–599	Reject	None

Why a scorecard first. A scorecard can be read line by line by a credit officer, an auditor and a supervisor. The boosted-tree challenger usually ranks risk a little better; it replaces the scorecard only after validation shows a real gain and its explanations hold up.

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5 REQUIREMENT 5 OF 8

Documents and property

Asked: verify documents digitally and assess the property within the same flow.

Document AI pipeline

1. **Capture:** in-app camera with edge detection, or PDF upload; DigiLocker documents come in already signed by the issuer.

2. **Quality check:** blur, glare and cropping are caught on the phone before upload.
3. **Classify:** the model tells a salary slip from a Form 16, a sale deed from an allotment letter.
4. **Extract:** fields such as name, PAN, employer, gross and net pay, survey number and owner are read, each with a confidence.
5. **Verify:** each field is checked against a source, as in the table below.
6. **Tamper check:** file metadata, font consistency, pixel patterns and QR codes are checked for edits.
7. **Store:** the file goes to the encrypted vault with a SHA-256 hash that the decision record references.

Document	Checked against
PAN card	PAN verification API
Aadhaar	UIDAI eKYC, offline XML signature or DigiLocker
Salary slips	Bank salary credits (AA) and EPFO contributions
Form 16 or ITR	AIS / Form 26AS
Bank statement PDF	AA data where possible; balance continuity otherwise
GST returns	GSTN
Sale agreement, allotment letter	RERA record and builder data
Title deed, mother deeds	State land records, encumbrance certificate, CERSAI
Approved plan, occupancy certificate	Municipal records where online
Property tax receipt	Municipal tax portal

Property assessment in the same flow

Identify	Address, survey or CTS number and map pin; for flats, the RERA project ID, tower and unit. If the project is pre-approved, most checks below are already done.
Ownership	The owner in state land records must match the seller (HL) or the applicant (LAP). Mutation status is checked.
Encumbrance	Encumbrance certificate from the state registration department, CERSAI search for an existing mortgage, and e-Courts search for cases.
Legal reading	The LLM deed reader pulls parties, survey numbers, dates and each transfer in the ownership chain, in English or regional languages. Every fact must point to text in the deed. Gaps or conflicts become flags.

Valuation	The AVM gives a value range from registered prices, circle rates, listings, building age and location. The circle rate acts as a floor check. If the declared price is far above the AVM, the case is flagged.
Inspection	For low-value or pre-approved cases, the customer records geo-tagged, time-stamped photos and a short video in the app. Otherwise an empanelled valuer does it through the vendor app. Photos are checked against satellite images and for reuse.
Technical	Approved plan, occupancy or completion certificate, building age, and flood or hazard zone from public maps.
Output	Market value, the value used for LTV (the lower of AVM and valuer value), and a legal status: clear needs review defective.

Where this stops being instant. Land records are digitised unevenly across states. Where a record is not machine-readable, the engine refers the title to an empanelled lawyer through the vendor portal. The rest of the file stays digital and the SLA is tracked.

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6 REQUIREMENT 6 OF 8

Compliance checks

Asked: show how NHB regulations, the Fair Practices Code and audit standards are built into the engine.

Each regulation becomes a **compliance rule** with an ID, the source clause, a version and an effective date. The compliance gate runs every rule before a decision leaves the engine. If a rule fails, the decision is blocked, the case goes to Refer, and the failure is logged. Each rule also lists the evidence it saves, so the audit trail fills itself.

Requirement	Source	How the engine enforces it	Evidence saved
LTV caps on individual housing loans (90% / 80% / 75%)	RBI HFC Directions	Hard rule; the loan amount is capped automatically	LTV maths, valuation ID
Customer due diligence: identity, address, CKYCR, V-CIP, risk category, periodic updates	RBI KYC Directions, 2025	No decision unless KYC status is verified	KYC artefacts, CKYC number, face and liveness scores
Enhanced checks and senior approval for politically exposed persons	RBI KYC Directions	Always Refer	Screening result, approver ID

Requirement	Source	How the engine enforces it	Evidence saved
Sanctions screening before onboarding; records kept 5 years after the relationship ends; suspicious transaction reports; no tipping-off	PMLA 2002, PML Rules 2005	Screening at onboarding and on list updates; retention tags; STR draft for the Principal Officer; neutral customer messages	Screening log with list version
Acknowledge each application; decide within a stated time; give rejection reasons in writing; terms in a language the borrower understands	Fair Practices Code (RBI HFC Directions, Chapter X)	Auto-acknowledgement with application ID; SLA timer; rejection letter built from reason codes; letters in the customer's language	Message log, timestamps
Share the reasons when a credit request is rejected	RBI rules on credit information (2023)	Reason codes always mapped to customer text	Letter copy
Key Facts Statement with all-in cost (APR) before the agreement	RBI KFS rules (2024); Digital Lending Directions, 2025	eSign is blocked until the KFS is shown and accepted	KFS PDF hash, acceptance time
Digital lending: need-based data with consent, cooling-off period, disbursal only to permitted accounts, grievance officer shown, app reported to RBI	RBI Digital Lending Directions, 2025	Consent manager; cooling-off flag on the loan; disbursal account check; grievance details in app; app listed on RBI's CIMS portal	Consent artefacts, disbursal account match
Penal charges instead of penal interest, with no capitalisation	RBI fair lending rules (2023)	Pricing engine has no penal-interest field; penal charges shown in the KFS	Pricing version
No foreclosure charges on floating-rate loans to individuals	RBI / NHB rules	Pricing rule	Pricing version
Interest rate by a board-approved model that grades risk	Fair Practices Code	Price spread comes from the risk grade table	Grade, rate table version
Register the mortgage with CERSAI within 30 days of creation	SARFAESI Act, 2002	Post-sanction task created with a due date	CERSAI filing ID
Return property documents within 30 days of full repayment	RBI rules on release of property documents (2023)	Closure task with a due date in the loan system	Release date
Consent, purpose limits, data minimisation, erasure when the purpose ends	DPDP Act 2023, DPDP Rules 2025	Consent ledger; retention engine; masking	Consent records, deletion log
IT governance, logs kept 180 days, cyber incidents reported within 6 hours	RBI IT Governance Directions (2023); CERT-In Directions (2022)	Central logging with retention; incident runbook	Log archive, incident tickets

Requirement	Source	How the engine enforces it	Evidence saved
Early warning signals and red-flagging for fraud	RBI Fraud Risk Management Directions (2024)	Fraud engine raises early warning signals; fraud register	Alert history
Responsible AI: board-approved policy, human oversight, explainability, audits	RBI FREE-AI framework (2025, advisory)	Model inventory, Refer path, reason codes, kill switch	Model cards, validation reports

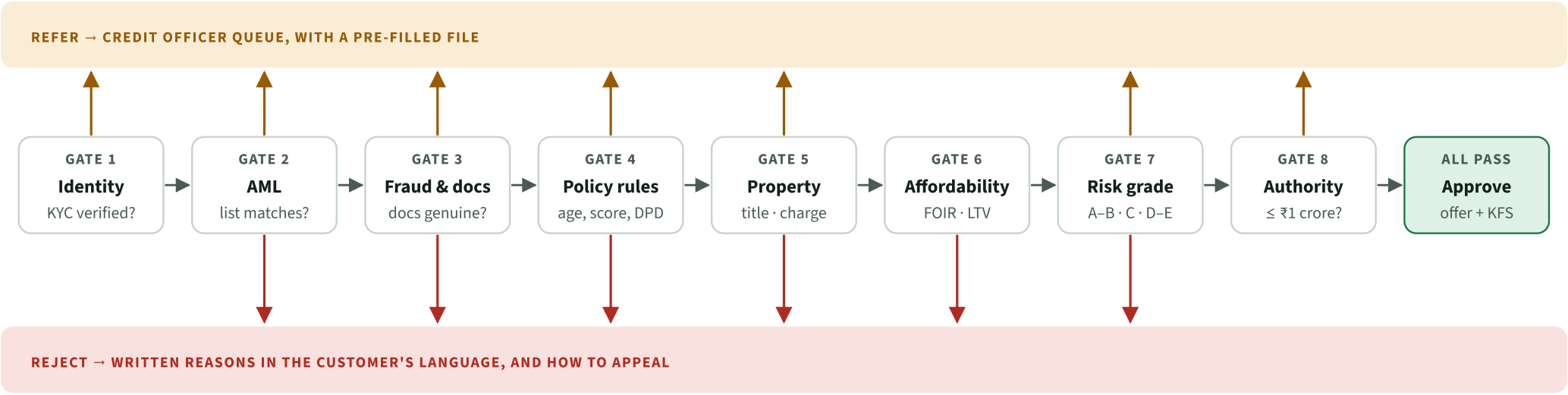
Audit standards built in

- **Maker-checker** on every rule, model and template change, with both names stored.
- **Tamper-evident trail:** each decision record includes the hash of the one before it, so a changed or deleted record breaks the chain. Records are kept on write-once storage.
- **Read-only audit portal** for internal audit, statutory auditors and NHB inspection teams, with replay of any decision.
- **Risk-based internal audit:** the audit portal gives the sample selection and exception reports that RBI's risk-based internal audit rules expect of larger HFCs.
- **Regulatory returns** are generated from the same data, so reported numbers match the engine.
- **Regulation changes** are handled in the policy studio: update the rule, cite the new clause, test on past cases, approve, then switch on at the effective date.

7 REQUIREMENT 7 OF 8

Instant decisioning

Asked: deliver a humanless credit decision instantly, with a clear approve, reject or refer outcome.



Every gate always runs, so every reason is recorded, not only the first. The outcome is the most severe result across the gates: any Reject means Reject, otherwise any Refer means Refer, otherwise Approve.

Gate	Refer when	Reject when
1 Identity	Face match or KYC data fails; video KYC offered	None
2 AML	Possible watch-list match; politically exposed person	Confirmed sanctions-list match
3 Fraud and documents	Income on documents over 30% above bank data; suspicious links	Issuer confirms a document was altered
4 Policy rules	Score 650–699; a 30+ day late payment; no credit history	Under 21; loan runs past maximum age; score below 650; a 90+ day late payment
5 Property	Title needs a lawyer	Undisclosed charge, or the seller is not the owner
6 Affordability	None; a smaller amount becomes a **counter-offer**	No room for an EMI, or eligible amount below half of the amount asked
7 Risk grade	Grade C	Grade D or E
8 Authority limit	Amount above ₹1 crore	None

What each outcome does

Approve

In-principle or final sanction letter with the KFS, the rate and EMI options, then eSign and the repayment mandate. A **counter-offer** is an approval for a lower amount or a different tenure, with the reason shown.

Refer

The case lands in the credit console with a file note, evidence links and the engine's draft offer. The officer must record a reason. Their decision is saved to the decision record and becomes a training label.

Reject

A written letter in the customer's language with the reasons (AML reasons excepted), what they can do next, such as adding a co-applicant, and how to ask for a review or raise a grievance.

Speed budget for one decision

Step	Target (p95)
Read features from the feature store	50 ms
Run policy and compliance rules	100 ms
Score models (credit, fraud)	150 ms
Build reasons and the decision record; write to the ledger	200 ms
Generate the letter and KFS	1,000 ms
Total	**about 1.5 s**

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8 REQUIREMENT 8 OF 8

Explainability and governance

Asked: every decision must be explainable, traceable and audit-ready.

Three audiences, three views of the same decision

Customer

Plain reasons in their language, what they can do next, and how to appeal. AML reasons are never shown.

Credit officer and auditor

Every gate's result, the rule that fired with its source clause, the points per factor, and links to the raw evidence.

Supervisor and model validator

Replay of any decision with the original inputs and the rule and model versions of that day, plus model cards and validation reports.

Reason codes

Every refer or reject reason has a code. Each code has an internal meaning and an approved customer message in each language.

Code	Internal meaning	What the customer is told
R-AFF-01	Existing EMIs use the full FOIR limit	Your current EMIs already use the maximum share of income we can lend against.
R-BUR-02	30+ days past due in the last 12 months	Your credit report shows a payment more than 30 days late in the last year.
R-PRP-01	Title needs legal review	Our lawyer needs to review the property title.
R-FRD-02	Document income does not match bank credits	The income on your salary slip does not match your bank credits. Please share your latest Form 16, or let us fetch it with your consent.
R-AML-01	Possible watch-list match	Your application needs some more review. We will contact you within one working day.
R-AUT-01	Above the auto-sanction limit	Loans of this size get a final check by a senior officer.

The decision record

Every decision writes one record to the audit ledger. It holds the inputs (or their evidence hashes), the version of every rule set and model used, each gate's result, the score with its points, the reason codes, the outcome, any officer action, and the hash of the previous record. Feeding the same inputs into the same versions gives the same output, so any decision can be replayed exactly. The simulator below builds a sample record live.

Governance

- **Board-approved policies** for credit, model risk and AI, with named owners.
- **AI and model risk committee** that approves every model and every material rule change.
- **Model inventory** with a model card for each model: purpose, data, limits, performance, fairness results, owner.
- **Independent validation** before go-live and at least once a year.
- **Monitoring:** drift, performance on matured loans, approval and refer rates by segment, and how often officers overturn the engine.
- **Fairness:** no protected attributes; proxy tests; approval-rate gaps reviewed by the committee.
- **Kill switch:** one action routes every new case to Refer.

Mapped to RBI's FREE-AI framework

RBI published the FREE-AI committee report on 13 August 2025, with seven guiding principles (“sutras”), six pillars and 26 recommendations. The recommendations are advisory until RBI issues directions, but they show what supervisors will look for.

Sutra	How Nirnay meets it
Trust is the foundation	Tamper-evident records, replay, evidence kept for every input
People first	Refer path, appeal route, plain-language reasons, help in Indian languages
Innovation over restraint	Challenger models in shadow, safe canary releases
Fairness and equity	No protected attributes, proxy tests, segment monitoring
Accountability	Named owners, maker-checker, committee approvals, officer reasons logged
Understandable by design	Scorecard as the live model, reason codes, SHAP for the challenger
Safety, resilience and sustainability	Guardrails, confidence thresholds, kill switch, DR site, drift monitors

Try it: decision simulator

This runs the decision logic from this report with the illustrative policy values in Requirement 2. Change any input and the outcome, the gate results, the score and the decision record update. It runs in your browser and sends nothing anywhere. In this PDF the simulator shows the Ravi example. Open the online report to try other inputs.

Product: Home Loan

Employment: Salaried

Age (years): 32

Net monthly income (₹): 100000

Existing EMIs (₹ a month): 10000

Bureau score: 765

☐ No credit history (new to credit)

Late payments in last 12 months: None

Property value (₹): 6000000

Loan asked (₹): 4500000

Tenure (years): 20

Interest rate (% a year): 8.75

CHECK RESULTS FROM OTHER SERVICES

KYC and face match | AML screening | Documents | Property title

Verified	Clear	Genuine, consister	Clear

Approve

₹45 lakh for 20 years at 8.75% · EMI ₹39,767 · Home Loan

Eligible amount	EMI	Total EMIs / income	Loan to value
₹45 lakh	**₹39,767**	**49.8%**	**75.0%**
limited by the amount asked	20 years at 8.75%	limit 50.0%	limit 80.0%

Risk score
738 · B
base 650

Gate by gate

1 Identity	Pass	eKYC, PAN and face match verified.
2 AML	Pass	No match on sanctions, UAPA or PEP lists.
3 Fraud & documents	Pass	Documents match source data; no tamper signs.
4 Policy rules	Pass	Age, tenure and bureau rules met (score 765).
5 Property	Pass	Title clear online; no existing charge in CERSAI or the EC.
6 Affordability	Pass	EMI ₹39,767 keeps total EMIs at 49.8% of income (limit 50.0%); loan is 75.0% of value (limit 80.0%).
7 Risk grade	Pass	Score 738, grade B.
8 Authority limit	Pass	Within the ₹1 crore auto-sanction limit.

How the score was built (points from base 650)

Repayment history		+33
	bureau score 765 against a 700 reference	
Recent late payments		+10
	none in 12 months	
Affordability		+10
	EMIs take 49.8% of income	
Collateral		+15
	loan is 75.0% of property value	
Stability		+20
	salaried income	
Product		0
	home purchase	
Total	650 + points, kept between 300 and 900	**738**

What the customer sees

- Your loan is approved in principle for ₹45 lakh with an EMI of ₹39,767 for 20 years.
- Next: review the Key Facts Statement, then sign digitally.

Decision record (sample)


```
{
  "record_id": "DR-DEMO-0001",
  "created_at": "2026-11-05T05:51:14.178Z",
  "product": "HL",
  "outcome": "APPROVE",
  "amount": 4500000,
  "emi": 39767,
  "tenure_years": 20,
  "rate_pct": 8.75,
  "policy_version": "HL-POL-2026.11-v1",
  "model_version": "RISK-SCORECARD-1.0-demo",
  "risk": {
    "score": 738,
    "grade": "B",
    "base": 650,
    "points": {
      "Repayment history": 33,
      "Recent late payments": 10,
      "Affordability": 10,
      "Collateral": 15,
      "Stability": 20,
      "Product": 0
    }
  },
  "gates": [
    {
      "gate": "Identity",
      "status": "pass"
    },
    {
      "gate": "AML",
      "status": "pass"
    },
    {
      "gate": "Fraud & documents",
      "status": "pass"
    },
    {
      "gate": "Policy rules",
      "status": "pass"
    },
    {
      "gate": "Property",
      "status": "pass"
    },
    {
      "gate": "Affordability",
      "status": "pass"
    },
    {
      "gate": "Risk grade",
      "status": "pass"
    },
    {
      "gate": "Authority limit",
      "status": "pass"
    }
  ],
  "reason_codes": [],
  "inputs": {
    "product": "HL",
    "emp": "salaried",
    "age": 32,
```

```
    "income": 100000,
    "emi": 10000,
    "bureau": 765,
    "ntc": false,
    "dpd": "0",
    "pv": 6000000,
    "ask": 4500000,
    "tenure": 20,
    "rate": 8.75,
    "kyc": "ok",
    "aml": "clear",
    "docs": "clean",
    "title": "clear"
  },
  "prev_record_hash": "(hash of the previous record in the ledger)",
  "record_hash": "sha256:0674d60122b2f6ee1e40314c7bef9c664f620e5b21e9d6e34062640d055b2022"
}
```


Hackathon build plan and team roles

The prototype should prove the whole flow end to end, with real logic and clearly labelled mock data where real access needs licences.

What we build

Piece	In the prototype	Real or mocked
Mobile app	Flutter: sign-in, consent, selfie, document upload, property photos, decision screen in English and Hindi	Real
Web portal	React: credit console with Refer queue, decision replay, policy studio with maker-checker	Real
Orchestrator and rules	FastAPI services, state machine, decision tables for HL and LAP	Real
Outside data	Bureau, AA, PAN, CKYCR, land records, CERSAI with realistic sample responses; AA sandbox if access is granted	Mocked
Document AI	OCR plus field rules and tamper checks on sample documents	Real
Risk model	Scorecard and LightGBM with SHAP, trained on the public Home Credit dataset	Real, demo data
Fraud and AML	Rules, a small link graph, and name matching on a sample list	Real, synthetic data
LLM deed reader	Reads three sample deeds; each fact must point to its source text	Real
Decision record	Hash-chained ledger and replay	Real

Who does what

Teams can have one to four members. With fewer than four, merge roles 1 and 3, then 2 and 4.

Role	Owns	Delivers for the demo
1 · Product and compliance lead	Policy rules, compliance table, KFS and letter templates, reason codes, the pitch	Rule tables, compliance matrix, demo script, slides
2 · App and portal	Flutter app, React portal, Hindi strings, accessibility	Six app screens, three portal screens
3 · Backend and integrations	Orchestrator, gateway, mock connectors, decision record, ledger	APIs, mocks, replay
4 · AI and ML	Document AI, risk model, fraud graph, AML matching, deed reader	Models with SHAP and short evaluation notes

Build order

1. Agree the data model, the eight gates and the reason codes as a team. This report is the starting point.
2. Build mock connectors and the rules engine; use the simulator's cases as the first tests.
3. Plug the AI pieces in behind the same service interfaces.
4. Build the app and portal on top of the APIs.
5. Add the decision record, replay and the audit portal.
6. Run the three demo cases end to end and rehearse.

Demo story

Case	Setup	Expected outcome	What it shows
Ravi, 32, salaried	HL ₹45 lakh on a ₹60 lakh flat in a pre-approved project	**Approve** in both stages within minutes	Straight-through processing, KFS, eSign
Meena, 45, runs a garment shop	LAP ₹60 lakh on a ₹1.2 crore property; deed has a gap in the ownership chain	**Refer** to legal via the vendor portal	Deed reader, Refer queue, officer decision logged
Arjun, 29, salaried	HL ₹30 lakh; salary slip was edited	**Reject** with reason and fraud flag	Tamper check, explainable rejection

The people in these cases are fictional.

Risks and limits

Risk	What we do about it
Real API access needs agreements and licences (Aadhaar KUA, bureau membership, AA registration)	Prototype uses mocks and sandboxes, clearly labelled; production goes through licensed partners
Land records are digitised unevenly across states	Refer to an empanelled lawyer through the vendor portal; pre-approved projects reduce the load
Model bias or hidden proxies	Excluded attributes, proxy tests, segment monitoring, committee review
LLM makes up facts	Every fact must point to source text; the LLM never decides; letters come from approved templates
No Hero training data in the hackathon	Public dataset plus expert scorecard, stated openly; retrain on Hero's book later
Rules change (RBI consolidated directions in November 2025 and has amended them since)	Each rule cites its clause and has an effective date; the compliance owner updates it in the policy studio
An outside source goes down	Circuit breakers and fallbacks; the step waits rather than guessing
Too much automation for edge cases	Authority limits, Refer path, override monitoring, kill switch
Data breach	Encryption, tokenised Aadhaar, least privilege, CERT-In reporting, DPDP breach process

Glossary

AA Account Aggregator: RBI-licensed network that moves financial data with the customer's consent.

APR Annual Percentage Rate: the all-in yearly cost of a loan, shown in the KFS.

AVM Automated Valuation Model: estimates property value from data.

CERSAI Central Registry of Securitisation Asset Reconstruction and Security Interest of India: records mortgages and runs the CKYC Registry.

CKYCR Central KYC Records Registry: one KYC record reused across lenders.

DPD Days Past Due: how late a payment was.

DSA Direct Selling Agent: a partner who brings loan customers.

EC Encumbrance Certificate: shows registered transactions and charges on a property.

FOIR Fixed Obligation to Income Ratio: total EMIs as a share of income.

FPC Fair Practices Code: conduct rules for how lenders treat borrowers.

GBM	Gradient-Boosted Machine: a strong tree-based ML model.
HFC	Housing Finance Company.
KFS	Key Facts Statement: a standard one-page summary of the loan's cost and terms.
LAP	Loan Against Property.
LTV	Loan-to-Value: loan amount as a share of property value.
PD	Probability of Default.
PEP	Politically Exposed Person.
RERA	Real Estate Regulatory Authority: registers real-estate projects in each state.
SHAP	A method that shows how much each input pushed a model's score up or down.
STR	Suspicious Transaction Report, filed with FIU-IND.
V-CIP	Video-based Customer Identification Process (video KYC).
WoE	Weight of Evidence: a way to turn input ranges into scorecard points.

Sources

The challenge rules ask us to cite any data or source we use. Regulatory points are summarised as of November 2026. RBI folded many 2023–2024 circulars into consolidated directions on 28 November 2025, so check the current text before submission.

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Policy values marked “illustrative” are our assumptions, not Hero Housing Finance's real credit policy. Timings are design targets, not measured results.

Nirnay · team solution report for Problem Statement 2, Hero Housing Fintech Challenge · Version 1.0, 5 November 2026